2026-06-10-SP4-cli-bridge-plan

SP4 — CLI-agent bridge providers — Implementation Plan

Field	Value
Revision	1
Created	2026-06-10
Last modified	2026-06-10
Status	active
Maintainer	CLI-Agent Fusion programme
Phase	PLANNING (read-only; no code modified to produce this plan)
Parent roadmap	`docs/superpowers/specs/2026-06-10-llms-access-master-roadmap.md` §SP4 (lines 154-167)
Source analysis	`docs/superpowers/specs/analysis/2026-06-10-D-cli-bridge.md`
Status summary	Decomposition complete; 34 tasks across P4.0–P4.6; tier-1 = 8 installed agents
Issues	docs/Issues.md (ATM-NNN assigned at execution kickoff)
Fixed	none yet

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1. Goal & scope

Make each working CLI coding agent its **own HelixCode model provider** that: 1. proxies prompts via the agent's CLI (real `exec`, never simulated), 2. surfaces power-features (Vision / Generative / RAG / Memory / MCP / LSP / ACP), 3. selects PRIMARY (system-installed binary via `exec.LookPath`) with FALLBACK (submodule-built binary), 4. lists models DYNAMICALLY per agent (CONST-036, no hardcoded lists), 5. re-exports + **installs** + **live-validates** each agent's config.

Tier-1 (do first — 8 agents system-installed NOW, versions captured in analysis-D §2): `claude` 2.1.170 → `qwen` 0.5.0 → `opencode` 1.16.2 → `gemini` 0.1.9 → `crush` v0.22.2 → `codex` → `goose` 1.8.0 → `copilot` 1.0.46.

Audio-priority ordering (§11.4.72) is N/A here (no audio surface); tier-1 ordering follows install-readiness + most-visible-first (§11.4.42 / §11.4.103).

Tier-2 (fallback path required — system-absent today): `aider`, `plandex`, `forge` (analysis-D §2). Then expand toward the ~43 coding-CLI candidates in `cli_agents/` (analysis-D §1).

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2. Verified anchors (read-only this session)

Every line below was opened and confirmed this session (not copied from analysis).

2.1 The `llm.Provider` contract HelixCode providers must satisfy

`helix_code/internal/llm/missing_types.go:356-378` — interface `Provider` with methods: `GetType() ProviderType`, `GetName() string`, `GetModels() []ModelInfo`, `GetCapabilities() []ModelCapability`, `Generate(ctx, *LLMRequest) (*LLMResponse, error)`, `GenerateStream(ctx, *LLMRequest, chan<- LLMResponse) error`, `IsAvailable(ctx) bool`, `GetHealth(ctx) (*ProviderHealth, error)`, `Close() error`, `GetContextWindow() int`, `CountTokens(text) (int, error)`. `ProviderType` const block at `missing_types.go:38+` already has `ProviderTypeQwen`, `ProviderTypeCopilot` etc. — new per-CLI provider types must be ADDED here.

2.2 `copilot_provider.go` is NOT a CLI bridge

`grep "exec.Command|LookPath" helix_code/internal/llm/copilot_provider.go` → **empty**. It is HTTP/token-based (confirmed). There is **no CLI-exec provider in** `helix_code/internal/llm` **today**.

2.3 D-7 wiring gap CONFIRMED

`helix_code/go.mod` replace block (lines 194-212) wires `containers/helixqa/docprocessor/llmorchestrator/visionengine/challenges/security/debate/helixspecifier/lazy` — **NO** `replace dev.helix.agent`. HelixCode cannot import the `clis` substrate today.

2.4 The real exec primitive — honest

`submodules/helix_agent/internal/clis/agents/base/base.go:120-124` `IsAvailable()` does real `exec.LookPath(string(b.info.Type))`; `:127-138` `ExecuteCommand` runs `exec.CommandContext` with workdir + env. This is the genuine detect+exec base to reuse. `agents/claude_code/claude_code.go` is partially real: `:160` `exec.LookPath("node")`, `:276` `exec.CommandContext(ctx,"bash",...)`, `:310` git exec, `:453` LookPath.

2.5 D-6 stub bluff CONFIRMED — and it is WIDER than the roadmap recorded

- `agents/qwencode/qwencode.go:101-112` `generate()` returns `fmt.Sprintf("// Generated by Qwen\n// %s", prompt)`; `:115-126` `chat()` returns `fmt.Sprintf("Qwen: %s", message)`; `:137-139` `IsAvailable()` returns `q.config.APIKey != ""` — **never execs** `qwen`. (Matches D-6 exactly.)
- **NEW FINDING (not in roadmap D-6):** the pooled dispatch methods in `submodules/helix_agent/internal/clis/instance_manager.go` are ALSO stubs — `executeAider:906-913`, `executeClaudeCode:915-922`, `executeCodex:924-931`, `executeCline:933-940`, `executeOpenHands:942+` each `return map[string]string{"status":"executed","message":"<Agent> execution completed", ...}` with **NO exec at all**. So even the "real CLI-exec layer" the analysis credits has a stubbed dispatch tier. This materially expands P4.0 de-bluff scope and MUST be RED-reproduced. (`SendRequest:290`, `CreateInstance:83`, `AcquireInstance:167` are the lifecycle entry points routing into these stubs.)
- The richer contract: `clis/types.go:537` `CLIAgent` interface (`ExecuteTask:548`), `Capabilities` struct `:555-566` (`SupportsRepoMap/GitOps/DiffEditing/ToolUse/Browser/Sandbox/Interpreter/SupportedTools`), `DefaultCapabilities:567+` hardcodes per-agent caps (CONST-040 says these SHOULD come from LLMsVerifier, not hardcode).

2.6 Re-export + static-validate EXISTS; filesystem-install + live-validate ABSENT

- `submodules/helix_agent/cmd/helixagent/main.go:107` flag `--generate-all-agents`, `:108` `--all-agents-output-dir`, dispatch `:1371` → handler `handleGenerateAllAgents:4522` (requires output dir `:4529`).
- `submodules/helix_agent/challenges/codebase/go_files/unified_cli_generator/unified_cli_generator.go`: `GenerateAll():76`, `saveConfig():212`, `Validate(agentType, config):263` → `*ValidationResult{Valid,Warnings,Errors}` (static validation only).
- `cli_agents_configs/` = **109 files** (`<agent>.json`/`<agent>.yaml` pairs + README). Their direction is REVERSED (CLI-agent → HelixAgent OpenAI-compat backend, analysis-D §4). SP4 adds the FORWARD direction (HelixCode → CLI-agent) and the ABSENT install+live-validate legs.

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3. Architecture decision: replace + thin adapter (CONST-051 / §11.4.74)

Decision (recommended, pending operator confirmation — see §9 D-A): add `replace dev.helix.agent => ../submodules/helix_agent` to `helix_code/go.mod` and write ONE thin adapter `helix_code/internal/llm/cli_agent_provider.go` that wraps the `clis` layer behind `llm.Provider`. Do **not** re-implement exec in HelixCode (§11.4.74 extend-don't-reimplement; CONST-051(B) decoupling — the generic bridge stays project-agnostic in the submodule, HelixCode owns only the thin adapter + registry).

Parametrized, not one-file-per-agent. A single `CLIAgentProvider` struct carries an `AgentSpec` (binary name, non-interactive arg template, model-list subcommand, capability probe set, fallback build path). Adding an agent = a registry row, not a new Go file. `GetType()` returns a per-agent `ProviderType` added to `missing_types.go`.

CONST-051(C) layout check: `dev.helix.agent` already lives at the root-accessible `submodules/helix_agent/` (flat layout). No nested own-org chain is introduced — the replace points at the existing root path. ✅

Module-boundary risk (flag to executor): `dev.helix.agent` is a large module (`internal/{agentic,ensemble,llm,mcp,memory,…}` — 40+ packages). Importing it transitively pulls its `go.mod` graph into `helix_code`. P4.0.2 MUST verify `go build ./...` still resolves after the replace and that no version conflicts arise; if the graph is too heavy, the fallback is a **narrow extraction** (a small `clis`-only interface shim) — decided at P4.0.2 with captured `go mod graph` evidence, not guessed (§11.4.6).

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4. Anti-bluff & TDD discipline binding this plan

- **§11.4.115 RED-first, polarity switch.** Every fix ships a RED test authored to FAIL on the CURRENT artifact (`RED_MODE=1` reproduces the stub bluff), flipping to GREEN guard (`RED_MODE=0`) post-fix. The canonical RED here: a test asserting `qwencode` (and each instance_manager dispatch method) *actually execs the binary* — it MUST fail on the current stub returning `"Qwen: %s"` / `"…execution completed"`.
- **§11.4.98 fully-automated.** Every governed test is self-driving end-to-end, no human action after startup, PASS at `-count=3`. Proxied-prompt tests drive a real installed CLI with a deterministic prompt (e.g. arithmetic) and assert a real answer in stdout — NOT a simulated string.
- **§11.4.108 four-layer.** SOURCE (provider code) → ARTIFACT (`go build` binary contains it) → RUNTIME-ON-CLEAN (provider registered + `IsAvailable` true on a clean checkout) → USER-VISIBLE (real proxied prompt → real result). Each agent declares ONE runtime signature (e.g. "`helixcode` CLI lists ≥1 model for `claude` via `GetModels` AND a `2+2` prompt returns `4`").
- **§11.4.135 regression guard** registered per closed defect in the same commit.
- **§11.4.85 stress + chaos** on the exec path (concurrent proxied prompts; binary-missing / timeout / killed-subprocess injection — must error cleanly, never crash).
- **§11.4.125 / §11.4.134 code-review gate** before pre-build + main build, iterate-until-GO.
- **§11.4.70 subagent-driven** execution default; one subagent per agent-bridge keeps context isolated.
- **CONST-036** dynamic `GetModels`; **CONST-040** capabilities from LLMsVerifier `VerificationResult` (not `DefaultCapabilities` hardcode) — wire to the

verifier surface SP1/SP2 own; where the verifier has no entry for a CLI agent yet, probe the agent at runtime and mark provenance honestly.

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5. Phased tasks

P4.0 — Pre-req remediation (anti-bluff, BLOCKS everything)

- **T4.0.1** RED-reproduce the `qwencode` stub bluff: test in `agents/qwencode/qwencode_test.go` asserting `generate/chat` run the real `qwen` binary and return non-templated output; assert `IsAvailable` uses `exec.LookPath`, not `APIKey != ""`. MUST FAIL on current code (§11.4.115).
- **T4.0.2** RED-reproduce the **instance_manager dispatch stubs** (NEW): test asserting `SendRequest→executeClaudeCode/executeCodex/executeAider/…` actually exec the binary; MUST FAIL on the current `"…execution completed"` map. (This is the wider bluff surface found §2.5.)
- **T4.0.3** GREEN: rewrite `qwencode` (`generate/chat/complete/IsAvailable/status`) over `base.ExecuteCommand` with the real `qwen` non-interactive invocation (research item §8). Keep package project-agnostic (CONST-051).
- **T4.0.4** GREEN: replace each stubbed `instance_manager.execute<Agent>` with a real dispatch into the agent integration's `Execute(...)` (which itself execs via `base`). Audit ALL ~70 agent packages for the same stub pattern; file a tracked sub-item per stubbed agent (do NOT silently leave others stubbed — §11.4.124 investigate-before-anything).
- **T4.0.5** Add `replace dev.helix.agent => ../submodules/helix_agent` to `helix_code/go.mod`; capture `go mod graph` + `go build ./...` evidence; decide full-import vs narrow-shim (§3 risk).
- **T4.0.6** §1.1 paired mutations for T4.0.1/T4.0.2 RED tests (re-insert a stub string → test FAILs).

P4.1 — Per-agent provider contract

- **T4.1.1** Add per-agent `ProviderType` consts to `missing_types.go` (e.g. `ProviderTypeClaudeCodeCLI`, …`QwenCodeCLI`, …`OpenCodeCLI`, …`GeminiCLI`, …`CrushCLI`, …`CodexCLI`, …`GooseCLI`, …`CopilotCLI`).
- **T4.1.2** `helix_code/internal/llm/cli_agent_provider.go` — `CLIAgentProvider` + `AgentSpec`, implementing all 11 `Provider` methods by delegating to `clis` (adapter). `IsAvailable` → real `exec.LookPath` mirror of `base.go:120`, never the qwencode APIKey bluff.
- **T4.1.3** `cli_agent_registry.go` — registry of `AgentSpec` rows (data, not code-per-agent); registration into the existing provider manager so `handleListModels` (root CLAUDE.md BLUFF-002 pattern) surfaces them.
- **T4.1.4** Unit + RED proxied-prompt test for the adapter against ONE installed agent (claude) before generalizing.

P4.2 — Primary-vs-fallback selection

- **T4.2.1** `resolveBinary(spec)` — PRIMARY `exec.LookPath(spec.Binary)`; FALLBACK resolve submodule-built artifact under `cli_agents/<agent>/` (built bin/dist/target or documented build target). Record chosen path in `GetHealth` details (anti-bluff evidence, analysis-D §5.3).

- **T4.2.2** Fallback build recipe wiring for aider/plandex/forge (tier-2). §11.4.77 regen-mechanism doc for each built binary.
- **T4.2.3** Tests: PRIMARY-present, PRIMARY-absent→FALLBACK-build, both-absent→honest `IsAvailable=false` + SKIP-with-reason (`hardware_not_present`-class → here `binary_unavailable`).

P4.3 — Power-feature passthrough + dynamic models + capabilities

- **T4.3.1** `GetModels()` per agent — exec the agent's model-list subcommand, parse, return `[]ModelInfo` (CONST-036). Research flags §8.
- **T4.3.2** `GetCapabilities()` — map `clis.Capabilities` → power-feature flags (Vision/Generative/RAG/Memory/MCP/LSP/ACP); source from LLMsVerifier `VerificationResult` where present (CONST-040), runtime-probe + honest-provenance otherwise.
- **T4.3.3** Power-feature passthrough plumbing (MCP/LSP/ACP/RAG/Memory) — pass HelixCode's MCP/LSP context into the agent invocation where the agent supports it; per-agent capability probe.
- **T4.3.4** `GenerateStream` line-buffered stdout → channel; `CountTokens/GetContextWindow` char-fallback (mirror copilot_provider char-estimate).

P4.4 — Config re-export + INSTALL + LIVE-validate (the ABSENT legs)

- **T4.4.1** Re-export: reuse `helixagent --generate-all-agents` + `unified_cli_generator.GenerateAll/saveConfig` (exists). Add forward-direction config (HelixCode → CLI-agent) generation.
- **T4.4.2 INSTALL (NEW):** copy generated config into each agent's real config dir (`~/.claude.json`, `~/.qwen/`, `~/.config/opencode/`, `~/.config/crush/`, etc.), back up existing first (§9.2 reversible-first). Per-agent install-path is a research item §8.
- **T4.4.3 LIVE-validate (NEW):** after install, run the installed agent end-to-end (real proxied prompt → real result, captured) to prove the install works (§11.4.5 / §11.4.69 runtime evidence). Static `Validate()` is the pre-check, live exec is the proof.
- **T4.4.4** `docs/qa/<run-id>/` transcript per agent (§11.4.83) — full bidirectional prompt→result.

P4.5 — Tier-1 rollout (8 installed) then expand

- **T4.5.1–T4.5.8** One subagent per tier-1 agent (claude, qwen, opencode, gemini, crush, codex, goose, copilot): wire `AgentSpec`, dynamic models, install+live-validate, capture evidence, register regression guard. Order = §1 tier-1 order.
- **T4.5.9** Expand-plan stub-item for tier-2 (aider/plandex/forge fallback build) + the ~43-agent backlog.

P4.6 — Full test coverage

- **T4.6.1** Per-agent real proxied-prompt e2e (§11.4.98 self-driving, `-count=3`).
- **T4.6.2** Stress + chaos on exec path (§11.4.85).
- **T4.6.3** HelixQA Challenge bank entry per tier-1 agent (CONST-050(B)).
- **T4.6.4** Pre-build gate `CM-CLI-AGENT-PROVIDER-NO-STUB` (greps provider + agent packages for templated-return bluff patterns) + §1.1 paired mutation.

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6. Files to create / modify

Path	Action	Purpose
`submodules/helix_agent/internal/clis/agents/qwencode/qwencode.go`	MODIFY	Replace stub `generate/chat/complete/IsAvailable/status` with real `exec` (T4.0.3)
`submodules/helix_agent/internal/clis/agents/qwencode/qwencode_test.go`	MODIFY	RED-on-stub → GREEN guard, `RED_MODE` switch (T4.0.1)
`submodules/helix_agent/internal/clis/instance_manager.go`	MODIFY	Replace stub `execute<Agent>:906+` dispatch with real integration dispatch (T4.0.4)
`submodules/helix_agent/internal/clis/instance_manager_test.go`	CREATE/ MODIFY	RED reproducing dispatch-stub bluff (T4.0.2)
`submodules/helix_agent/internal/clis/agents/*/` (audit ~70)	MODIFY (as found)	De-bluff each stubbed agent package; tracked sub-item each (T4.0.4)
`helix_code/go.mod`	MODIFY	Add `replace dev.helix.agent => ../submodules/helix_agent` (T4.0.5 / D-7)
`helix_code/internal/llm/missing_types.go`	MODIFY	Add per-agent `ProviderType` consts (T4.1.1)
`helix_code/internal/llm/cli_agent_provider.go`	CREATE	`CLIAgentProvider` adapter implementing `llm.Provider` (T4.1.2)
`helix_code/internal/llm/cli_agent_registry.go`	CREATE	`AgentSpec` registry + provider-manager registration (T4.1.3)
`helix_code/internal/llm/cli_agent_provider_test.go`	CREATE	Adapter unit + RED proxied-prompt tests (T4.1.4)
`helix_code/internal/llm/cli_agent_fallback.go`	CREATE	`resolveBinary` primary→submodule fallback (T4.2.1)
`helix_code/internal/llm/cli_agent_install.go`	CREATE	Filesystem install + live post-install validation loop (T4.4.2/T4.4.3)
`helix_code/tests/integration/cli_agent_bridge_test.go`	CREATE	Per-agent real proxied-prompt e2e, `-tags=integration` (T4.6.1)

Path	Action	Purpose
`helix_code/tests/integration/cli_agent_stress_chaos_test.go`	CREATE	Stress + chaos on exec path (T4.6.2)
`helix_code/scripts/gates/cm-cli-agent-provider-no-stub.sh`	CREATE	Anti-stub pre-build gate + paired mutation (T4.6.4)
`cli_agents_configs/forward/<agent>.{json,yaml}`	CREATE	Forward-direction (HelixCode→agent) configs (T4.4.1)
`docs/qa/<run-id>/`	CREATE	Per-agent install + proxied-prompt transcripts (T4.4.4)
`docs/Issues.md`, `docs/CONTINUATION.md`	MODIFY	ATM-NNN tracking + state sync (CONST-044)

(Inner-module paths are under the repo-root `helix_code/` tracked subdirectory; `cli_agents_configs/`, `submodules/`, `docs/` are repo-root.)

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7. Ordered task list (34 tasks)

1. T4.0.1 RED qwencode stub
2. T4.0.2 RED instance_manager dispatch stubs (NEW surface)
3. T4.0.3 GREEN qwencode real exec
4. T4.0.4 GREEN instance_manager dispatch + audit ~70 packages
5. T4.0.5 `replace dev.helix.agent` + `go build` graph evidence
6. T4.0.6 §1.1 paired mutations for P4.0
7. T4.1.1 per-agent ProviderType consts
8. T4.1.2 `CLIAgentProvider` adapter
9. T4.1.3 `AgentSpec` registry + manager registration
10. T4.1.4 adapter unit + RED proxied-prompt (claude)
11. T4.2.1 `resolveBinary` primary→fallback + health provenance
12. T4.2.2 fallback build recipe (aider/plandex/forge) + §11.4.77 regen doc
13. T4.2.3 selection tests (present/fallback/both-absent SKIP-with-reason)
14. T4.3.1 dynamic `GetModels` per agent (CONST-036)
15. T4.3.2 `GetCapabilities` from LLMsVerifier (CONST-040) + honest probe
16. T4.3.3 power-feature passthrough (MCP/LSP/ACP/RAG/Memory/Vision)
17. T4.3.4 `GenerateStream` + token-count fallback
18. T4.4.1 forward-direction re-export
19. T4.4.2 filesystem INSTALL with backup (ABSENT leg)
20. T4.4.3 LIVE post-install validate (ABSENT leg)
21. T4.4.4 `docs/qa/<run-id>/` transcripts
22. T4.5.1 tier-1: claude
23. T4.5.2 tier-1: qwen
24. T4.5.3 tier-1: opencode
25. T4.5.4 tier-1: gemini
26. T4.5.5 tier-1: crush
27. T4.5.6 tier-1: codex
28. T4.5.7 tier-1: goose
29. T4.5.8 tier-1: copilot
30. T4.5.9 expand-plan: tier-2 fallback + ~43-agent backlog
31. T4.6.1 per-agent real proxied-prompt e2e (§11.4.98, count=3)

32. T4.6.2 stress + chaos on exec path
33. T4.6.3 HelixQA Challenge bank per tier-1 agent
34. T4.6.4 `CM-CLI-AGENT-PROVIDER-NO-STUB` gate + paired mutation + code-review-until-GO

Dependency rule: P4.0 blocks all; P4.1 before P4.2/P4.3/P4.5; P4.4 needs P4.1+P4.2; P4.5 needs P4.1–P4.4; P4.6 cross-cuts P4.5. Within P4.5, tier-1 subagents run in parallel (§11.4.103) but each binary is a single-owner resource during its live-validate (§11.4.119).

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8. Per-agent §11.4.99 research items

Verify against LATEST official docs (never memory); cite URLs + date in a `## Sources verified` footer per agent before its config/guide commits. Each item = (a) non-interactive invocation, (b) model-list subcommand, (c) JSON/structured output mode, (d) real config-file path for INSTALL.

- **Claude Code 2.1.170** — `claude -p` / `--print` / `--output-format json` print mode; model selection flag; session reuse; MCP wiring; config path `~/.claude.json` (confirm).
- **Qwen Code 0.5.0** — real non-interactive invocation + model list (current `qwencode.go` is a stub, must be rebuilt against the installed binary); config dir `~/.qwen/` (confirm).
- **OpenCode 1.16.2** — `opencode run`? non-interactive flags; how "Zen" provider models are enumerated/selected; model-list command; config `~/.config/opencode/` (confirm).
- **Gemini CLI 0.1.9** — non-interactive flag, model-list subcommand, output mode.
- **Crush v0.22.2** — non-interactive flag, model list, `~/.config/crush/` (confirm).
- **codex** (no `--version` output) — confirm invocation + model list + approval-mode flags.
- **goose 1.8.0** — non-interactive flag, session/model selection, model list.
- **copilot (GitHub Copilot CLI) 1.0.46** — non-interactive flag, model list, auth handoff (distinct from the HTTP `copilot_provider.go`).
- **Power-features per agent** — which of Vision/RAG/Memory/MCP/LSP/ACP each exposes and how to probe at runtime (CONST-040).
- **Tier-2** — aider / plandex / forge build-from-submodule recipe + non-interactive invocation.

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9. Operator decisions required

- **D-A (architecture):** confirm `replace dev.helix.agent` + thin adapter (recommended) vs porting a `clis`-only shim into HelixCode. Surfaces at P4.0.5 with `go mod graph` evidence (§11.4.66 interactive options).
- **D-B (config install target):** INSTALL writes into the operator's real per-agent config dirs (`~/.claude.json`, `~/.qwen/`, …), overwriting/backing-up live config. This mutates operator state — confirm before T4.4.2 (reversible-first backup is mandatory regardless).
- **D-C (scope of de-bluff):** the ~70-agent stub audit (T4.0.4) may surface many stubbed packages beyond qwencode + instance_manager. Confirm: fix-all-now vs fix-tier-1-now + track-rest (§11.4.124 forbids silent deletion; all must be tracked).
- **D-D (no removal):** several `cli_agents_configs/` files point agents AT HelixAgent (reverse direction). SP4 ADDS forward configs — it MUST NOT remove the reverse ones without operator yes (§11.4.122).

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10. Rollback plan

- Each task is its own commit; the `replace` line + adapter are independently revertible.
- P4.0 submodule de-bluff lands in `dev.helix.agent` with its own RED→GREEN evidence; if the import graph proves too heavy (T4.0.5), revert the `helix_code/go.mod` replace and fall back to the narrow-shim path — the de-bluff itself stays (it fixes a live CONST-035 bluff regardless of wiring choice).
- INSTALL (T4.4.2) backs up every overwritten config before writing; rollback = restore the backup.
- No force-push (§11.4.113); all integration via merge-onto-latest-main.
- Gates reconcile-not-fake-pass (§11.4.120) if a P4.0 fix breaks an existing stub-tolerant test.

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11. 12-line summary

1. SP4 makes each working CLI agent its own HelixCode `llm.Provider` (contract `missing_types.go:356`), proxying prompts via real `exec` + surfacing Vision/RAG/Memory/MCP/LSP/ACP.
2. Confirmed D-7: `helix_code/go.mod` (194-212) has NO `replace dev.helix.agent` — wiring gap blocks importing the `clis` substrate.
3. Confirmed D-6 stub bluff: `qwencode.go:101/115/137` returns templated strings + gates on `APIKey!=""`, never execs `qwen`.
4. NEW finding beyond the roadmap: `instance_manager.go:906+` dispatch methods (`executeClaudeCode/executeCodex/executeAider/executeCline/executeOpenHands`) are ALSO stubs returning `"…execution completed"` — the “real exec layer” has a stubbed dispatch tier; P4.0 scope widened.
5. The honest exec primitive is `base.go:120` (`exec.LookPath`) + `:127` (`exec.CommandContext`); reuse it, don’t reimplement (§11.4.74).
6. Architecture: `replace dev.helix.agent` + ONE parametrized `CLIAgentProvider` adapter (CONST-051 decoupling) — adding an agent = a registry row, pending operator D-A.
7. Re-export + static `Validate():263` exist (`unified_cli_generator.go`, `main.go:4522`); the ABSENT legs are filesystem INSTALL + LIVE post-install validation — SP4’s net-new work.
8. `cli_agents_configs/` (109 files) point agents AT HelixAgent (reverse); SP4 adds the forward direction without removing reverse (§11.4.122).
9. TDD: every fix RED-first on the broken artifact with `RED_MODE` polarity switch (§11.4.115), fully-automated `-count=3` (§11.4.98), four-layer + runtime signature (§11.4.108).
10. Per-agent §11.4.99 research: OpenCode/Zen run+model-list, Claude Code `-p/--output-format json`, Qwen real invocation, plus gemini/crush/codex/goose/copilot non-interactive flags + config paths.
11. 34 tasks across P4.0 (de-bluff+wire) → P4.1 (contract) → P4.2 (fallback) → P4.3 (models+caps+power-features) → P4.4 (install+live-validate) → P4.5 (tier-1 rollout) → P4.6 (full coverage + gate).
12. **Task count: 34. Tier-1 order: claude → qwen → opencode → gemini → crush → codex → goose → copilot.**

Sources verified

2026-06-10 (read-only this session): `missing_types.go:356`, `copilot_provider.go` (no exec), `helix_code/go.mod:194-212`, `base/base.go:120-138`, `qwencode/`

`qwencode.go:101-139`, `instance_manager.go:82-181,290,906-942`, `types.go:537-567`, `unified_cli_generator.go:76,212,263`, `cmd/helixagent/main.go:107-108,1371,4522`, `claude_code/claude_code.go:160,276,310,453`, `cli_agents_configs/` (109 files). Per-agent CLI flag/version docs are §8 research items — NOT yet verified against vendor docs (must be before their config/guide commits, §11.4.99).